Radio, light signals, transponder

Edition Oct 2026

Tower light signals (AIM 4-3-13)

Signal	Aircraft in flight	Aircraft on the ground
Steady green	Cleared to land	Cleared for takeoff
Flashing green	Return for landing (steady green follows)	Cleared to taxi
Steady red	Give way to other aircraft and continue circling	Stop
Flashing red	Airport unsafe, do not land	Taxi clear of the runway in use
Flashing white	Not applicable	Return to starting point on airport
Alternating red and green	Exercise extreme caution	Exercise extreme caution

Vehicles and people: steady green go · steady red stop · flashing red clear the taxiway or runway · flashing white return to starting point.

Where a balloon needs what

Airspace	Transponder and ADS-B Out	Radio
A B C	Required unless ATC authorizes	B: ATC clearance first. C: two-way contact before entering.
Above a B or C ceiling, inside its limits, to 10,000 MSL	Required: no balloon exemption	
Mode C veil (30 NM of a listed Class B airport)	Exempt outside A, B, C and below the lower of the B or C ceiling or 10,000 MSL	
At and above 10,000 MSL, 48 states, not within 2,500 ft AGL	Exempt	
D	Not required (91.215, 91.225)	Two-way contact before entering

Exempt: balloons and aircraft certificated without an engine-driven electrical system (91.215(b)(3), (b)(5); 91.225(e)). Unequipped: request a deviation 1 h ahead. Contact is established when ATC answers with your call sign. Class D radio failure, VFR: land if basic VFR, tower in sight, cleared to land (91.129(d)(2)).

Squawk

1200 VFR · **7500** hijack
7600 lost radio · **7700** emergency

Calling

Facility, call sign, position, request: *"Palomar Tower, Balloon One Two Three Alfa Bravo, five miles east…"* Say the type and drop the N. **MAYDAY** ×3 distress, **PAN-PAN** ×3 urgency; emergency frequency **121.5**.

Alfa Bravo Charlie Delta Echo Foxtrot Golf Hotel India Juliett Kilo Lima Mike November Oscar Papa Quebec Romeo Sierra Tango Uniform Victor Whiskey Xray Yankee Zulu · 3 "tree" · 4 "fow-er" · 5 "fife" · 9 "niner"

Sources: AIM 4-3-13, AIM 4-1-20, AIM 6-2-2, AIM 6-3-1, AIM 6-3-4, AIM 6-4-2, AIM 4-2-3, AIM 4-2-4, AIM 4-2-7, AIM 3-2-4, AIM 3-2-5, 14 CFR 91.125, 14 CFR 91.129, 14 CFR 91.130, 14 CFR 91.131, 14 CFR 91.215, 14 CFR 91.225. Original card, not an FAA document. Study aid, not for navigation.